

Probability model

Sample space Ω + Prob map

$$P: \cancel{\mathcal{P}(\Omega)} \rightarrow \mathbb{R}$$

Σ -algebra: $\Sigma_{\Omega} \subseteq \mathcal{P}(\Omega)$

1) $\Omega \in \Sigma_{\Omega}$

2) $X: \mathbb{N} \rightarrow \Sigma_{\Omega}$ then $\bigcup_n X_n \in \Sigma_{\Omega}$

3) $X \in \Sigma_{\Omega} \Rightarrow X^c \in \Sigma_{\Omega}$

Prob map: $P: \Sigma_{\Omega} \rightarrow \mathbb{R}$

1) $P(\Omega) = 1$

2) $P\left(\biguplus_n X_n\right) = \sum_n P(X_n)$

3) $P(X) \geq 0$

Example: 1) X finite $(X, P(X), P(A \subseteq X) = \frac{\#A}{\#X})$

uniform distribution

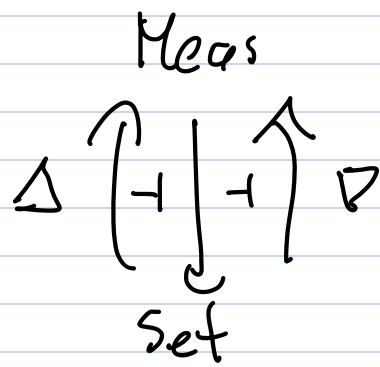
2) Borel alg: smallest Σ -alg. on $\mathbb{R}$ containing all $\mathcal{J}a, b[= (a, b)$

$$\text{Lebesgue measure} = P([a, b]) = b - a$$

Category of meas. set Meas

$f: (X, \Sigma_X) \rightarrow (Y, \Sigma_Y)$ is measurable

$$\text{if } \forall A \in \Sigma_Y \quad f^{-1}(A) \in \Sigma_X$$



$$\Delta(X) = (X, \mathcal{P}(X))$$

$$\nabla(X) = (X, \{\emptyset, X\})$$

$f: X \rightarrow Y$ always meas as map

$$f: (X, \mathcal{P}(X)) \rightarrow (Y, \Sigma_Y)$$

$$\text{Meas}(\Delta X, (Y, \Sigma_Y)) \cong \text{Set}(X, Y)$$

$$\text{Meas}(X, \Sigma_X, \nabla Y) \cong \text{Set}(X, Y)$$

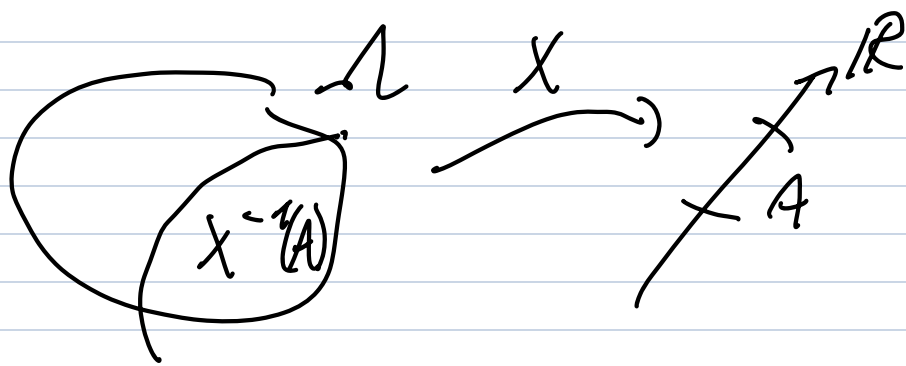
Random variable

$$X: (\Omega, \Sigma_\Omega) \rightarrow (\mathbb{R}, \mathcal{B})$$

If μ meas on $(\Lambda, \Sigma_\Lambda)$

$X_\#(\mu) : \text{Meas. on } (\mathbb{R}, \mathcal{B})$

$$X_\#(\mu)(A \in \mathcal{B}) = \mu(X^{-1}(A))$$



Lat of X

Product measure

$$(X, \Sigma_X) \otimes (Y, \Sigma_Y) = (X \times Y, \Sigma_{X \otimes Y})$$

$\Sigma_{X \otimes Y}$ is smallest Σ -alg. containing

all $A \times B \in X \times Y$ for $A \in \Sigma_X, B \in \Sigma_Y$

Construct product meas

$$\mu \otimes \nu(A \times B) = \mu(A) \times \nu(B)$$

Lebesgue integral:

If (X, Σ_X) & $f: (X, \Sigma_X) \rightarrow (\mathbb{R}, \mathcal{B})$

& suppose μ meas. on (X, Σ_X)

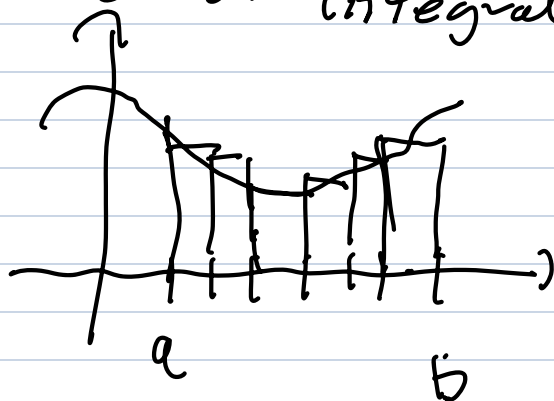
Define

$\int f d\mu = \lim$ over all partitions
of X into meas $B_1, \dots, B_n$,

pick $x_i \in B_i$

$$\sum_{i=1}^n f(x_i) \cdot \mu(B_i)$$

Riemann integral



$$\int f d\mu = \int_{x \in X} f(x) \mu(dx)$$

Markov kernels:

Usual
 $f: X \rightarrow \mathcal{M}_Y$

Actual def

$$P: X \rightarrow \Sigma_Y \rightarrow \mathbb{R}$$

1) $\forall x \ P_x$ meas. on $\mathcal{Y}$

2) $\forall A \in \Sigma_{\mathcal{Y}} \ P(-)A: X \rightarrow \mathbb{R}$ is measurable

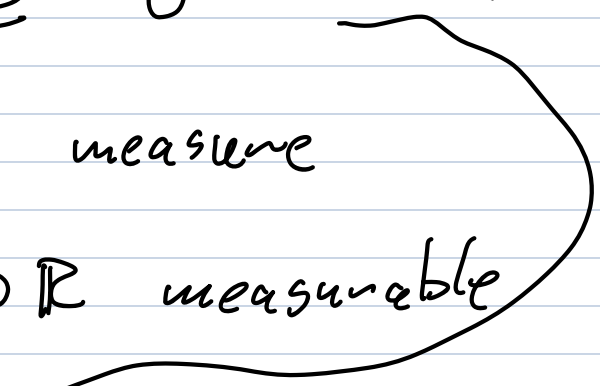
If P markov kernel $X \rightarrow \mathcal{Y}$
 $Q \text{ --- } \parallel \text{ --- } \mathcal{Y} \rightarrow \mathcal{Z}$

$(P; Q) \text{ --- } \parallel \text{ --- } X \rightarrow \mathcal{Z}$

$$(P; Q) \times (A \in \Sigma_{\mathcal{Z}}) = \int Q(-, A) (P_x)$$

$P_x: \Sigma_{\mathcal{Y}} \rightarrow \mathbb{R}$ measure

$Q(-, A): \mathcal{Y} \rightarrow \mathbb{R}$ measurable


$$\int_{y \in \mathcal{Y}} Q(y, A) P_x(dy)$$

Unit: $S: X \rightarrow \Sigma_X \rightarrow \mathbb{R}$

$$S_x A = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{else} \end{cases}$$

Next time: Idea is

$f: X \rightarrow Y$ is a probabilistic prog.

Model it as

$$[f]: [X] \rightarrow M[Y]$$

Essentially a Markov kernel

In chapter model

$$[f]: M[X] \rightarrow M[Y]$$

$$\begin{array}{ccc} [X] & \xrightarrow{f} & M[Y] \\ \downarrow \delta & \nearrow & \\ M[X] & \xrightarrow{f_*} & \end{array}$$

Use $P: X \rightarrow \Sigma_Y \rightarrow \mathbb{R}$ Markov kernel

Define $P_0: M X \rightarrow M Y$

$$P_{\mu} \mu (A \in \Sigma_Y) = \int P(-, A) \mu \in \mathbb{R}$$

Define $M X$ to be all meas.

$\Sigma_X \rightarrow \mathbb{R}$ including neg. one

vector space. In next week

$T: M_X \rightarrow M_Y$ which is linear